

ABHINAV PANDEY

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EDUCATION

Bharatiya Vidya Bhavans Sardar Patel Institute of Technology

June 2026

B.Tech Electronics and Telecommunications

Current GPA: 8.3 / 10.0

Matushri Kashiben Vrajlal Valia International Vidyalaya

June 2022

XII

Percentage: 93.4/100

SKILLS

Languages: C/C++, JavaScript/TypeScript, Python, SQL

Backend & Tools: Node.js, Express, FastAPI, PostgreSQL, Git/Github, Postman, Swagger, Cmake

Cloud: Firebase, Supabase, DigitalOcean Droplets, GCP

Other: Figma, Unix/Linux

PROJECTS

Rest-O-Route | *Python, fastapi, Elasticsearch, Selenium, Here Maps API* | 

May 2025

- Built the backend for a multi-stop journey planning app, improving trip personalization based on dietary and accessibility needs.
- Automated scraping of 500+ restaurant entries using Selenium, evaluating hygiene ratings, allergen compliance, and safety violations to create a health-aware dining database.
- Designed a custom routing algorithm that dynamically adjusted travel paths according to accommodation needs, reducing planning time in usability tests.
- Developed a full-text search system using Elasticsearch, achieving sub-second query responses and 90% accuracy in relevance-ranked restaurant suggestions.

Smart Waste Management System | *Micropython, GCP, ClearBlade IoT Core, expo* | 

Nov. 2024

- Developed an end-to-end IoT-based waste collection system using ClearBlade IoT Core for MQTT-based communication between microcontroller-equipped bin sensors and the cloud.
- Designed an event-driven backend pipeline with Google Cloud Pub/Sub and Cloud Functions to handle real-time sensor alerts and trigger optimized collection workflows.
- Integrated Google Cloud SQL to store bin metadata, alert history, and collection logs, enabling robust monitoring and analytics.
- Co-developed a cross-platform mobile app using Expo to deliver location-based alerts to waste collection drivers, reducing response time, achieving timely and efficient waste management.

Virtual Stock Market (VSM) | *Typescript, Node.js, Supabase, Express, PostgreSQL* | 

Jan. 2024 - Feb. 2024

- Developed the backend for a gamified stock market simulator used by 200+ students during the Oculus 2024 college festival.
- Modeled real-time trading mechanics based on actual market behavior, improving user engagement and retention by 40% over previous year's static model.
- Built a secure login and trading system using Supabase Auth and PostgreSQL, supporting over 1,000 transactions/hour during peak usage.
- Containerized and deployed the system using Docker on DigitalOcean, achieving 99.9% uptime during the 3-day event.

COMPETITIONS & ACHIEVEMENTS

HERE Technologies Hackathon 2025

Secured position in the Top 5 by making a product using HERE APIs.

SPIT SE Hackathon 2024

Secured finalist position among 200+ participating teams.

ROLES & RESPONSIBILITIES

EXTC Department Tech Team | *Member*

Sept. 2024 – May 2025

Served as a maintainer of the EXTC Department website. Worked alongside the department administration to update/improve the website before the department received government accreditation, ensuring that the latest and accurate data were available.